CIRC HQ STUDENT HANDOUT | GRADE 5

Crash Lander

Goal: Design a landing system that reduces impact on a fragile payload during a controlled drop.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 7, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Point to the impact evidence and explain how the revision changed the force path.

You will use

- Small payload containers
- Paper
- Cardboard
- Straws
- Tape
- Cotton
- Reusable test payloads
- Drop-zone team cards

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Count your allowed materials.
- Place the payload beside the container.
- Pick up one drop-zone team card.

2 Step 2: Drop zone safety — 3 min

- Stay behind the drop-zone line.
- Let the teacher release the lander.
- Do the waiting job on your team card.
- Move only when the teacher calls your team.

3 Step 3: Plan the lander — 5 min

- Mark where the payload needs support.
- Sketch one part that can bend safely.

4 Step 4: Build the capsule — 11 min

- Place the payload inside the container.
- Add only allowed materials.
- Check that no part can fall off.

5 Step 5: Test and revise — 7 min

- Predict the first failure point.
- Watch the controlled drop.
- Inspect the payload.
- Change one feature.

6 Step 6: Reset the drop — 3 min

- Return the reusable payload.
- Collect every loose part.

7 Step 7: Exit check — 3 min

- Draw the impact evidence.
- Label your useful change.

What to turn in

Impact evidence and one revision.

No computer? Another way to do it

The teacher does low drops from table height, or design on paper.

Finished early? Next challenge

Mass Limit: Reduce the capsule's material while keeping the payload protected in the same drop test.

Stay safe

Only the teacher releases from raised heights. Everyone stays behind the drop-zone boundary.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.